True Random Generators and Hidden Transfer of Keys

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***Abstract*—Most industrial automatic systems are remotely controlled, so the problem of protecting control information transmitted through an open channel is relevant. The use of cryptographic systems for security involves the ability to fast exchange system keys. In this paper, a new family of the true random key generators is proposed. The source of entropy for such random devices is the phenomenon of jittering in electronic circuits. Generators are built in the form of cascade schemes based on adders. We give the architectural solution for the synthesis of the generator that based on transition dynamics of such an appliance. The procedure for the hidden transfer of keys using those generators is discussed.**

Keywords—true random number generators, information security, key management

I. INTRODUCTION

Most industrial devices nowadays require access to the Internet. Companies not only sell the system once, but also accompany it in the process of using, remotely receiving data about the device, monitoring its condition, carrying out preventive work, or even full production management [1–3]. The transfer of information technologies to the automation domain will provide major opportunities.

On the other hand, the information became more vital so does the need for stronger security [4–6]. Security based on cryptographic techniques is commonly used in many automation applications from various fields [7,8]. If the data must be protected against an unauthorized access a password or a cryptographic system are leveraged. Both approaches must be supported by a generator of random values [9,10]. In the first case, the password is produced by a generator, in the second case, the system keys must be random numbers that are generated by a special device or a program. Also, such generators can be a part of a system intended for the hidden transfer of data. The true random number generators are preferred for those purposes.

Recently, a new family of such generators is under research. They are based on the notion jittering of the combinational circuit with feedback [11,12]. That phenomenon arises if a signal at the input of the combinational circuit depends on the outputs of the circuit.

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The simplest version of a suchlike generator is presented in Fig.1. Here, the inverter is used as a source of random numbers (jittering). SR - synchronous register that stores only those output signal of the inverter which arrives with synchrony impulse. The register is a mandatory part of any generator in our approach, so in what follows, we do not include the SR into the picture when we present the structure of a generator.

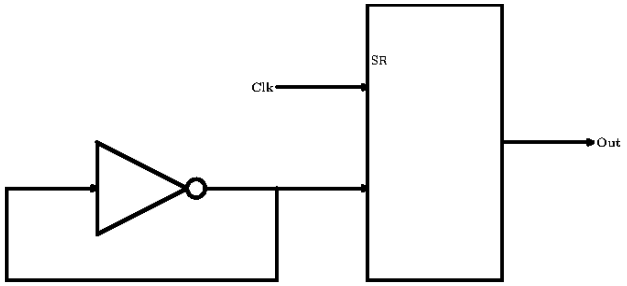


Fig. 1. Example of a generator that uses jittering phenomenon.

II. RELATED WORK

In this this section, we gather all definitions and the results that are pertained to the family of the generators under consideration. In our case, the combinational circuit, which works in non-standard mode, consists of modulo 2 adders over *GF(2)* [13]. Each gate has two inputs and one output. Some inputs of the gates can be free. It means that a constant signal is applied to that free input. To control the dynamics of the unit we use those signals at free inputs. There is a feedback that can convert the scheme either into a unit with memory or into a generator. An example of the unit is presented in Fig.2. The unit consists of 4 gates, and two of them have free input. In our previous papers, a theory of such schemes was developed [14].

A. Connection Matrix

The structure of a scheme can be specified in terms of a binary matrix, that is, consisting of ones and zeros. Let us suppose that there are N gates and M free inputs in the circuit. The connection matrix Conn is of size N×(N+M). Let Conn[i,j] be the item of the matrix from the i-th row and and the j-th column, Conn[I:J,P:Q] be submatrix of Conn, that is placed at the intersection of rows with numbers I,I+1,…,J-1 and columns with numbers P,P+1,…,Q-1, Conn[i,:] be the i-th row of the matrix. The analogous notation is used for access to components of vectors.

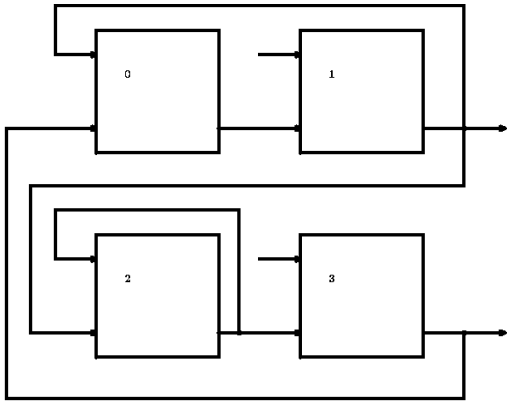


Fig. 2. Example of a scheme containing modulo 2 adders and feedback.

All the gates are numbered by integers from {0,1,...*N*-1}. The free inputs are numbered as 0,1,...,*M*-1. Matrix element *Conn[i,j]*=1, *j* < *N*, if and only if one of the inputs of the *i*th gate is connected with the output of the *j*th gate. Element *Conn[i,j]*=1, *j* ≥ *N* if and only if the free input with the number *j*-*N* is connected to one of the inputs of the *i*th gate. It means that the index of those free input is equal to *j*-*N*. For example, the connection matrix pertained to the scheme in Fig.2 has the form

$$Conn = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

It follows from the definition that in each row of the matrix Conn we have exactly 2 ones and all other elements are zeros. Each column of the matrix can include the arbitrary number of ones. Evidently, one can restore the scheme up to the numbering of gates and inputs using this matrix.

B. Dynamics

In [14], a model was proposed to describe the dynamics of the aforesaid device. It is based on Erlang equations for Markov process with continuous time. Recall basic facts related to that model.

Let α=<s1 ,s2 ,… , sN-1> be a vector whose components are outputs of all gates at some time t. We call this vector the state of the unit at time t. In what follows, the vector γ = <α,β> will be used where β=<x1, x2, … , xM-1> contains all signals on free inputs at that time. We denote this vector as an extended state of the scheme.

In accordance with the definition, the state of the unit can be changed only if one of the gates changes its output. Let us introduce a new matrix operation Θi:

$$\alpha'^{T} = Conn\ \Theta_i\ \gamma^T \qquad (1)$$

The meaning of such an operation is as follows: α′[i] = α[i], j ≠ i and α′[i]=Conn[i,*]·γT is a regular matrix product [15,16]. From the point of view of the gates, the adder with number i performs adding of the signals on its inputs. In fact, it is possible that the output of the gate will be the same as before the operation, but we consider this situation as if the adder fulfilled summation.

Applying the formula (1), one creates the transient matrix Tr_β of the size $K \times K$, $K=2^N$. This matrix depends on signals β on free inputs that assumed to be constant. All the states are numbered by integers from the interval $0,1,\ldots,2^N-1$. One to one correspondence between the state α and index m is set by the function $\varphi: \varphi(m) \leftrightarrow \alpha$. The transient matrix with a given vector β is built according to *Algorithm 1*.

Algorithm 1. Creating a transient matrix

1. Tr←K=2^N {create K×K zero matrix }
2. β ← β_0 { current vector containing values on free inputs}
3. FOR Q=0 TO K-1 DO
4. α ← φ (Q)
5. γ ← <α,β>
6. FOR i =0 TO N-1 DO
7. α′ ← Conn Θ_i γ^T
8. R ← φ^{-1} (α′)
9. Tr[R,Q] ← Tr[R,Q] +1
10. ENDFOR
11. ENDFOR

From *Algorithm 1* it follows that all items in matrix *Tr* are integers from the interval {0,1,...,*N*}. Since the sum of elements for any column in *Tr* equals to *N*, so Tr^T/N is a stochastic matrix [15].

Let $P_n(t)$ be the probability that at time *t* the scheme will be in the state with number *n*. It is supposed that at any time only one gate can change its output. Using standard reasoning performed during inference Erlang equations for queuing, one concludes that asymptotic behavior of the vector $\langle P_0(t), P_1(t), \ldots, P_{K-1}(t)\rangle$, depends on properties of matrix Tr_β [14].

III. MODULAR SCHEMES

As was shown in the previous section, the size of the transient matrix extends rapidly with the increase of the number of gates. On the other hand, while design a scheme, one often tries to build it from blocks of small sizes. In [17], the cascade structure of schemes was considered (Fig.3). Here F_0 and F_1 are schemes with feedback but there is no feedback from F_0 to F_1.

In [17] the procedure was presented to create a transient matrix of the scheme in the case we know transient matrices of blocks. In the above paper, we considered local feedback between blocks. In this paper, another case is investigated where global feedback between blocks exists (see Fig.4).

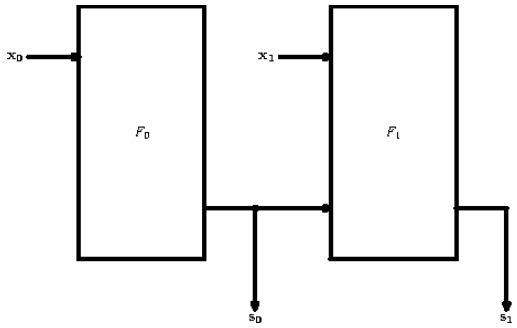


Fig. 3. Structure of cascade scheme.

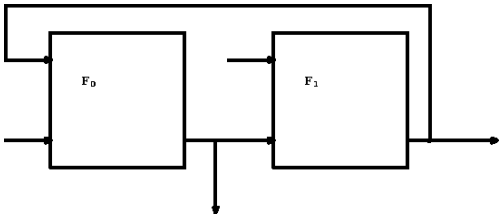


Fig. 4. Scheme with global feedback.

A. Connection Matrix

Let us suppose that both the schemes F_0 and F_1 are defined by their connection matrices *Conn0* and *Conn1*. To build matrix *Conn* pertained to the arisen scheme one has to point the connections between the schemes F_0 and F_1. We can do it by means of two lists $FromF_0toF_1$ and $FromF_1toF_0$. Each list contains records of the form: *IndexofOutput* and *IndexofFreeInput*. The method for calculation index of free input is presented in the definition of connection matrix. Let us demonstrate these notations by the scheme in Fig.2. That scheme can be built from two schemes in Fig.5 and Fig.6.

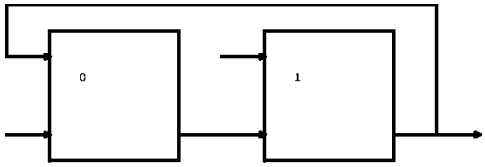


Fig. 5. Scheme F_0.

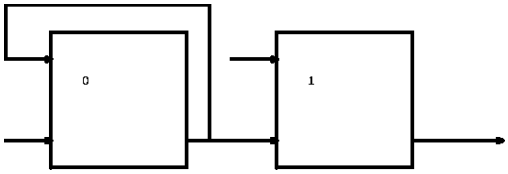


Fig. 6. Scheme F_1.

The connection matrices are as follows:

$$Conn0 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

$$Conn1 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

The lists, defining communications, have the form $FromF_0toF_1 = [[1,0]]$ and $FromF_1toF_0 = [[1,0]]$. The algorithm for the calculation of a connection matrix is presented in *Algorithm 2.*

Algorithm 2. Calculation of connection matrix on the base of connection matrices of F_0 and F_1

1. N0,Col0 ← Conn0 {size of matrix, N0 – number of gates}
2. N1,Col1← Conn
3. Ln0 ← $FromF_0toF_1$ {length of list}
4. Ln1 ←$FromF_1toF_0$
5. N ← N0+N1 {number of gates }
6. Col ← Col0 +Col1 –Ln0 – Ln1
7. Conn ← N, Col {zero matrix of size N×Col }
8. Conn[0:N0,0:N0] ← Conn0[:,0:N0]
9. Conn[N0:N,Col0:Col0+N1] ← Conn1[:,0:N1]
10. FOR Record in $FromF_0toF_1$ DO
11. Set connection pertained to Record
12. Delete free input from Conn0
13. ENDFOR
14. FOR Record in $FromF_1toF_0$ DO
15. Set connection pertained to Record
16. Delete free input from Conn1
17. ENDFOR
18. Add free inputs to Conn

After generating the matrix *Conn* according to *Algorithm 2*, there is a possibility to combine free inputs. To this end, one has to move all combined free inputs into one column of *Conn*, saving position in the column. For example, if in the scheme in Fig.2 both free inputs are combined then the connection matrix has the form

$$\mathrm{Conn} = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}. \quad (2)$$

Once the connection matrix is constructed, the transient matrix is generated by *Algorithm 1*.

B. Modular Scheme as Generator

If a device is intended to produce random values, it must not have stable states. A priori a device possibly cannot change its state. For example, suppose the equation holds

$$\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0 & 1 & 1\\1 & 0 & 1\end{pmatrix}\odot_i\begin{pmatrix}1\\0\\1\end{pmatrix} \quad (3)$$

for i=0,1. It means that the state <1,0> is stable if 1 is applied to the free input.

It is proved in [18] necessary and sufficient condition for the unit has no stable state. To formulate it, let us introduce some notations. Let *Ident* be the identity matrix, N be the number of gates, and β be the vector applied to free inputs. The units have no stable states if and only if

$$rank(Ident + Conn[:,0:N]) <$$

$$< rank((Ident + Conn[0:N,0:N]|)|\Delta), \quad (4)$$

where $\Delta = Conn[:, N:]\cdot\beta^{T}$ is a regular matrix product over *GF(2),* and vertical line denotes addition column to the matrix. Evidently, zero state is always stable if β is a zero vector. In what follows we exclude that situation. Condition (4) can be checked very easily in automatic mode while the design of the scheme. In [14] the following proposal was proved.

Theorem. The cascade scheme in Fig.3 where F_0 is an inverter, implemented as a feedback adder and the other input is set to 1, has no stable states.

For example, the circuit in Fig.4 has no stable states when both blocks are equivalent to blocks of type in Fig.6:

$$Conn=\begin{pmatrix}1 & 0 & 0 & 1 & 0 & 0\\1 & 0 & 0 & 0 & 1 & 0\\0 & 1 & 1 & 0 & 0 & 0\\0 & 0 & 1 & 0 & 0 & 1\end{pmatrix} \quad (5)$$

The next step in the study of the properties of the generator is obtaining transition diagram and searching the existence of the final distribution for the vector $<P_0(t),P_1(t),\ldots,P_{K-1}(t)>$, $t\to\infty$. If the single final distribution exists then it is the normalized eigenvector of the transient matrix Tr_β related to eigenvalue N. It is possible that the multiplicity of the eigenvalue N is more than 1. In this case, obtaining the final distribution requires additional consideration.

C. Implementation of Random Generators for Hidden Information Transfer

In this section, we present a procedure for hidden transfer data through an open channel. The main idea is well known. The information is transferred as a signal that looks like white noise. The recipient analyzes the signal by means of a known algorithm. Here we present a possible implementation of that idea. Let us explore the schema in Fig.7.

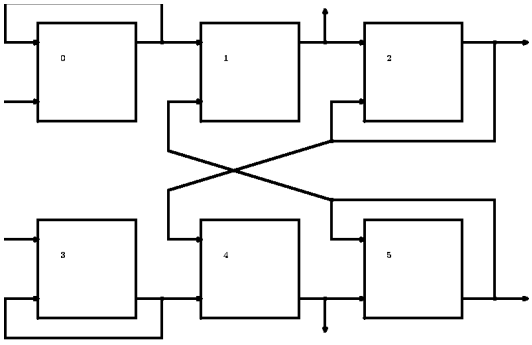


Fig. 7. Scheme for the realization of hidden data transfer.

That is a simplified version of the unit. One can extend the schematic solution for the case with an arbitrary number of gates. All the statements, that are analogous to ones formulated below, hold for those extended versions too. The scheme has two free inputs, and the output of the scheme consists of outputs of gates 1,2,4,5. The connection matrix has the form

$$Conn=\begin{pmatrix}1 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\1 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\0 & 1 & 1 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 1 & 0 & 0 & 0 & 1\\0 & 0 & 1 & 1 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 1 & 1 & 0 & 0\end{pmatrix}.$$

The admissible signals on the free inputs are (1,0), (0,1), and (1,1). According to the *Theorem*, the scheme has no stable states while one of the admissible signals is applied to free inputs.

Consider the situation when the input vector (1,0) is applied. In this case, the signal on the output of gate 3 will be constant. Thus the transition diagram of the generator contains at last two connected components. Direct calculation of matrix Tr_{10} spectra shows that the multiplicity of the eigenvalue 6 is 2. Consequently, we have exactly two connected components in the transition diagram. On the other hand, Tr_{11} has single eigenvector related to eigenvalue 6. All components of that vector are positive values. It means that the transition diagram has a single connected component and each state is attainable from any other state.

The idea of the hidden data transfer is as follows. Let us associate (1,0) with 1, (0,1) with 0, and (1,1) with space between two symbols in the source file. We denote these three signals as control signals of the generator.

One supposes that the interval between two clocks in the synchronous register (Fig.1) is a big enough to set the Markov process into states that are close to the final distribution. Let us compare the probabilities of various states in the final distribution while control signals are applied. In our case, the output vector has length 4.

First of all, let compare the probabilities when the vectors (1,0) and (0,1) are applied. Because of the symmetry of the scheme, output probabilities of (0,0,0,0), (0,1,1,0), (1,0,0,1), and (1,1,1,1) will be the same.The differences are depicted in Fig.8.

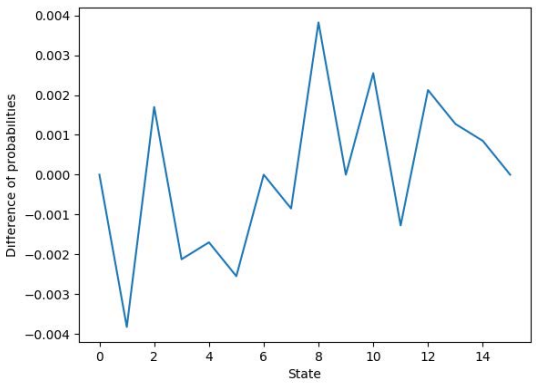


Fig. 8. Differences of probabilities of various states for (0,1) and (1,0) control signals.

It follows from the Fig.8 that the most significant differences are gained for vectors $1 = (0,0,0,1)$ and $8 = (1,0,0,0)$, so using these two signals as markers one can distinguish control signals and bits of information.

To select the space between two bits one has to compare the probabilities where (0,1), (1,0), and (1,1) are applied to the free inputs. Those differences are presented in Fig.9.

It follows From Fig.8 and Fig.9 that variety of signals can be leveraged to detect a control signal. A number of clocks (length of random sequence), used to transmit one bit with given reliability, is a standard problem known as hypothesis testing [19,20].

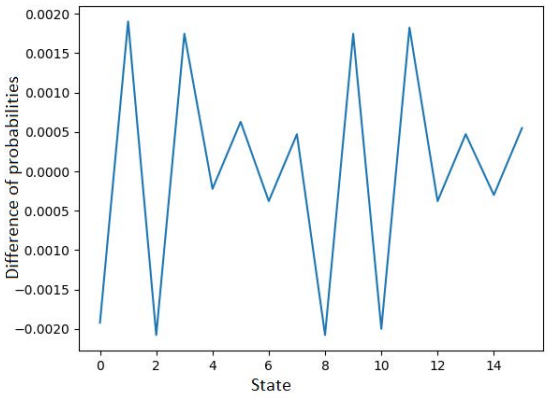


Fig. 9. Differences in probabilities for (1,0) and (1,1) are applied for the realization of hidden data transfer.

IV. Conclusion

A new family of true of true random number generators as cascade (module) circuits has developed. Theoretical research, as well as experiments, show the suitability of generator for hidden data transfer.

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